UNIT – IV

Fatigue Behaviour & Testing

Subject : Mechanical Behaviour of Materials (23A03504c)

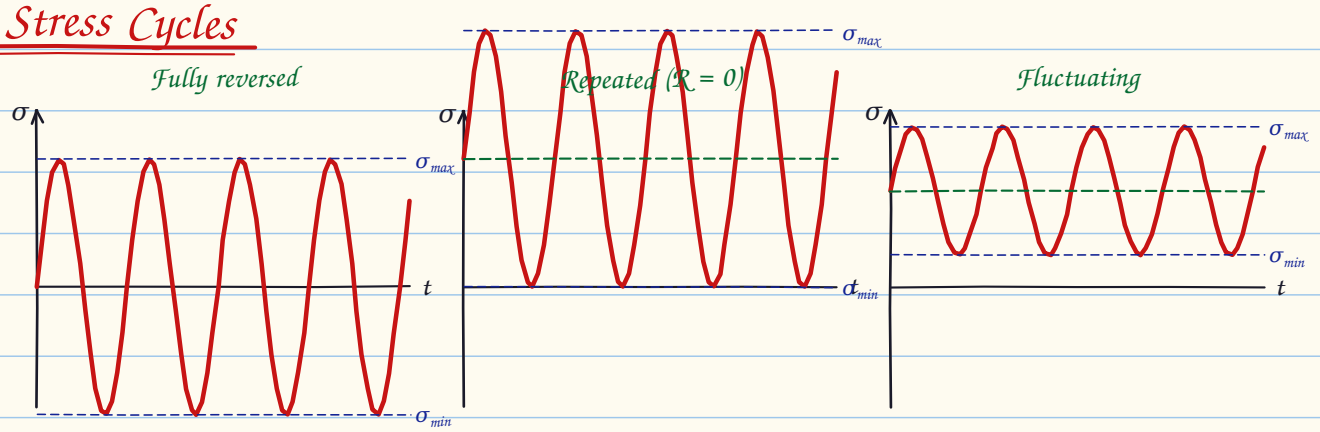
Syllabus: Stress cycles, S–N curves, effect of mean stress, factors affecting fatigue, structural changes accompanying fatigue, cumulative damage, HCF/LCF, thermo-mechanical fatigue, fracture mechanics applied to fatigue crack propagation (Paris law), fatigue testing machines.

1. Introduction – Fatigue

Fatigue = progressive, localised, permanent structural change in a material subjected to fluctuating stresses and strains that may culminate in cracks or complete fracture after a sufficient number of cycles – at stress levels well below the static UTS (often below yield).

- It accounts for ≈ 90 % of service failures of metallic parts (shafts, crankshafts, springs, aircraft wings, bridges, rails, gears, turbine blades, welded joints). Examples : Comet airliner (1954) window corners ; Versailles rail accident 1842 (axle) ; Alexander Kielland platform 1980.
- Characteristics : failure without warning ; crack starts at the surface (stress raisers) ; fracture surface has two zones – smooth fatigue zone with beach marks, and a rough final fast-fracture zone.
- Fatigue life $N_f = N_i$ (initiation) + N_p (propagation) – in HCF initiation dominates (≈ 90 %), in LCF propagation.

2. Stress Cycles



$\sigma_m = (\sigma_{max}+\sigma_{min})/2$; $\sigma_a = (\sigma_{max}-\sigma_{min})/2 = \Delta\sigma/2$; $R = \sigma_{min}/\sigma_{max}$; $A = \sigma_a/\sigma_m$

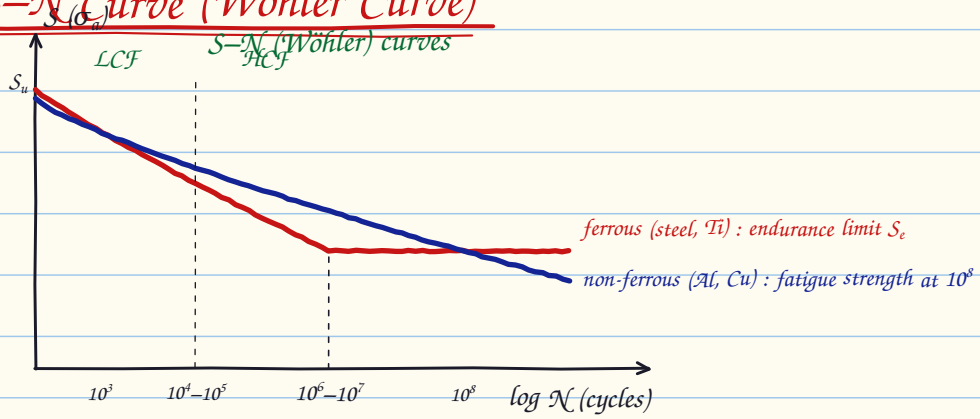
- **Stress range** $\Delta\sigma = \sigma_{max} - \sigma_{min}$. **Stress amplitude** $\sigma_a = \Delta\sigma/2$. **Mean stress** $\sigma_m = (\sigma_{max} + \sigma_{min})/2$. **Stress ratio** $R = \sigma_{min}/\sigma_{max}$; **amplitude ratio** $A = \sigma_a/\sigma_m = (1 - R)/(1 + R)$.

$$\sigma_{max} = \sigma_m + \sigma_a \quad \sigma_{min} = \sigma_m - \sigma_a$$

- Types : *fully reversed* ($\sigma_m = 0$, $R = -1$ – rotating shafts, R.R. Moore) ; *repeated / pulsating* ($\sigma_{min} = 0$, $R = 0$ – gear teeth, springs) ; *fluctuating* ($\sigma_m \neq 0$, general) ; random / spectrum (aircraft gust loads) ; block loading.

Ex 1. $\sigma_{max} = 300$ MPa, $\sigma_{min} = -100$ MPa : $\sigma_m = 100$ MPa ; $\sigma_a = 200$ MPa ; $\Delta\sigma = 400$ MPa ; $R = -100/300 = -0.33$.

3. S–N Curve (Wöhler Curve)



Plot of stress amplitude S (σ_a) versus the number of cycles to failure N (log scale), obtained by testing a set of identical specimens at different stress levels (usually $R = -1$).

- **Ferrous metals (steel, Ti)** : curve becomes horizontal at ≈ 10^6–10^7 cycles – the endurance (fatigue) limit S_e : the stress below which failure never occurs ; ≈ 0.4–0.5 S_u for steels ($S_e' = 0.5\ S_u$ for $S_u \le 1400$ MPa, capped at 700 MPa). Due to strain-ageing / locking of dislocations by interstitials (C, N).
- **Non-ferrous (Al, Cu, Mg) and aged steels** : no true limit ; the curve continues to fall – use fatigue strength at a specified life (10^7–5×10^8 cycles). $S_e \approx 0.4\ S_u$ at 5×10^8 for Al alloys.
- **Regions** : LCF ($N < 10^3$–10^5, plastic), HCF (10^5–10^7, elastic), infinite life. Basquin's law in the HCF region :

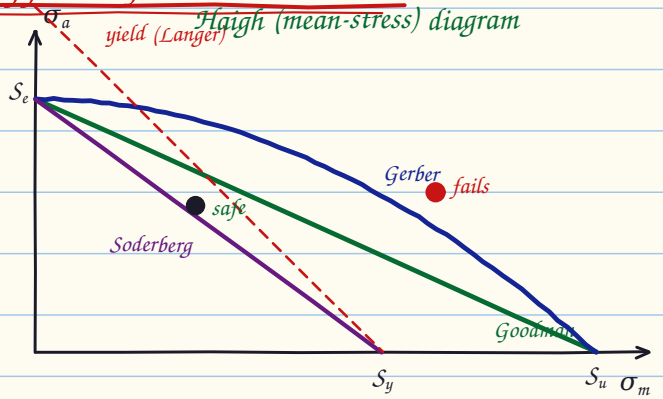
$$\sigma_a = \sigma_f' (2N_f)^b \quad (b = -0.05 \text{ to } -0.12 \ ; \ \sigma_f' \approx \text{true fracture strength})$$

- Scatter : ± factor 2–10 in life for the same stress – statistics (log-normal / Weibull) ; S–N–P curves ; **staircase method** to find S_e (≈ 15–30 specimens).

Ex 2. $\sigma_f' = 1000$ MPa, $b = -0.1$: for $\sigma_a = 300$ MPa : $2N = (0.3)^{-10} = 1.69\times10^5$ ⇒ $N = 8.5\times10^4$ cycles.

4. Effect of Mean Stress

Haigh (mean-stress) diagram



Goodman : $\sigma_a/S_e + \sigma_m/S_u = 1$
Gerber : $\sigma_a/S_e + (\sigma_m/S_u)^2 = 1$
Soderberg : $\sigma_a/S_e + \sigma_m/S_y = 1$
Compressive σ_m : little effect

A tensile mean stress reduces the allowable stress amplitude ; a compressive mean stress is beneficial. Empirical relations (Haigh diagram) – σ_a for infinite life at a given σ_m :

$$\text{Goodman} : \sigma_a/S_e + \sigma_m/S_u = 1 \quad \text{Gerber} : \sigma_a/S_e + (\sigma_m/S_u)^2 = 1 \quad \text{Soderberg} : \sigma_a/S_e + \sigma_m/S_y = 1$$

- **Goodman** – straight line, conservative for brittle materials (best fits tests) ; **Gerber** – parabola, fits ductile data ; **Soderberg** – most conservative, protects against yielding ; **Langer** – yield line $\sigma_a + \sigma_m = S_y$. Others : Morrow,

Smith–Watson–Topper (SWT : $\sigma_{max}\varepsilon_a E$).

- Factor of safety n : Goodman $1/n = \sigma_a/S_e + \sigma_m/S_u$. Compressive mean stress – assume $\sigma_m = 0$ (or small benefit).

> Ex 3. $S_e = 250$ MPa, $S_u = 600$ MPa, $S_y = 400$ MPa, $\sigma_m = 100$ MPa : Goodman $\sigma_a = 250(1 - 100/600) =$ 208 MPa ; Gerber $\sigma_a = 250(1 - 0.0278) =$ 243 MPa ; Soderberg $\sigma_a = 250(1 - 0.25) =$ 187.5 MPa.

> Ex 4. Component with $\sigma_{max} = 300$, $\sigma_{min} = 100$ MPa : $\sigma_m = 200$, $\sigma_a = 100$ MPa ⇒ Goodman $1/n = 100/250 + 200/600 = 0.733$ ⇒ $n =$ 1.36.

5. Factors Affecting Fatigue

The laboratory endurance limit S_e' is modified (Marin factors) for the real component :

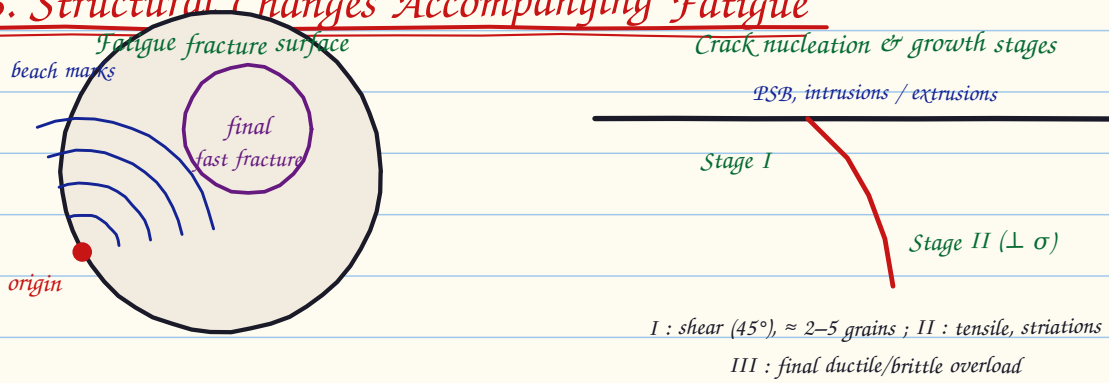
$S_e = k_a k_b k_c k_d k_e k_f S_e'$ $k_a = a S_u^{\,b}$ (surface) ; $k_b = 1.24\, d^{-0.107}$ (size, 2.8–51 mm) ; $k_c = 1$ (bending), 0.85 (axial), 0.59 (torsion) ; k_d (temperature) ; k_e (reliability 0.897 for 90 %, 0.814 for 99 %) ; k_f miscellaneous

- **1. Stress concentration** (holes, notches, keyways, threads, fillets, sharp corners, welds) : fatigue notch factor $K_f = 1 + q\,(K_t - 1)$ (q = notch sensitivity 0–1, rises with notch radius and material strength) ; S_e (notched) $= S_e/K_f$. Remedy : large fillet radii, smooth transitions, stress-relief grooves, avoid abrupt section changes.
- **2. Surface finish** : polished surfaces best ; machined, hot-rolled, forged reduce S_e (scratches = micro-notches, decarburisation). k_a (machined) $= 4.51\, S_u^{-0.265}$.
- **3. Size effect** : larger section ⇒ larger stressed volume ⇒ more flaws, steeper gradient ⇒ lower S_e.
- **4. Residual stresses** : compressive surface stresses (shot peening, cold rolling of fillets, burnishing, carburising, nitriding, induction hardening, autofrettage) increase fatigue life 20–100 % (retard crack initiation) ; tensile residual stresses (welding, grinding, quenching) reduce it. Removed by high T / overload.
- **5. Environment** : corrosion fatigue – synergy of cyclic stress and corrosive medium ⇒ no endurance limit, S–N curve keeps falling ; fretting fatigue at contact surfaces (press fits, bolts) – loses 30–70 % ; hydrogen ; high T (creep–fatigue).
- **6. Metallurgical** : grain size (fine grain ↑ S_e), inclusions/porosity (casting defects, ↓), banding, anisotropy (transverse < longitudinal), decarburised layer, microstructure (tempered martensite best), hardness ($S_e \approx 1.7$–1.8 HB up to 400 HB). **7. Temperature, frequency, overloads, mean stress, type of loading, surface**

treatments (chromium plating reduces S_e), coatings.

Ex 5. Machined steel shaft $d = 20$ mm, $S_u = 600$ MPa, bending : $S_e' = 300$ MPa ; $k_a = 4.51(600)^{-0.265} = 0.83$; $k_b = 1.24(20)^{-0.107} = 0.90$; $S_e = 300 \times 0.83 \times 0.90 = 224$ MPa. With a shoulder fillet $K_t = 2.5$, $q = 0.8$: $K_f = 1 + 0.8 \times 1.5 = 2.2$ ⇒ allowable amplitude $224/2.2 \approx 102$ MPa.

6. Structural Changes Accompanying Fatigue



Stages of fatigue failure

- **Stage I – crack initiation** : repeated cyclic slip on the most favourably oriented systems forms persistent slip bands (PSBs) (concentrated slip, dislocation ladder structure in Cu : veins and walls) ; slip steps at the surface form **extrusions** (ribbons pushed out) and **intrusions** (micro-notches) – the nucleus of the crack, at grain boundaries, inclusions, notches, in about the first 10 % of life. Crack grows along the slip plane (≈ 45° to σ) through 2–5 grains – **Stage I growth** (shear mode).
- **Stage II – crack propagation** : the crack turns normal to the maximum tensile stress and advances by blunting and re-sharpening at the tip each cycle, leaving **striations** (one per cycle, spacing ≈ da/dN, 0.01–1 μm, visible in the SEM) ; larger **beach (clamshell) marks** mark changes in loading or interruptions, seen by naked eye, concentric around the origin.
- **Stage III – final fracture** : when the remaining cross-section can no longer carry the load ($K_{max} = K_c$) ⇒ sudden ductile (tear) or brittle fracture. Relative area of the fast-fracture zone ⇒ indicates the stress level (small zone ⇒ low stress, large fatigue zone).

Cyclic deformation

- **Cyclic hardening** (annealed metals – dislocation multiplication) or **cyclic softening** (cold-worked/hard metals – dislocations rearrange) until a saturated hysteresis loop is reached ; the **cyclic stress–strain curve** (locus of loop tips) differs from the monotonic curve : cyclic hardening if $n' > n$. Indicator : if $S_u/S_y > 1.4$ hardens, < 1.2 softens.
- **Bauschinger effect,** hysteresis-loop energy per cycle ≈ plastic damage ; at high temperature cracks become intergranular (creep–fatigue).
- **Fatigue crack closure** (Elber) : plasticity-induced residual wake closes the crack before zero load ⇒ effective $\Delta K_{eff} = K_{max} - K_{op}$.

7. Cumulative Damage

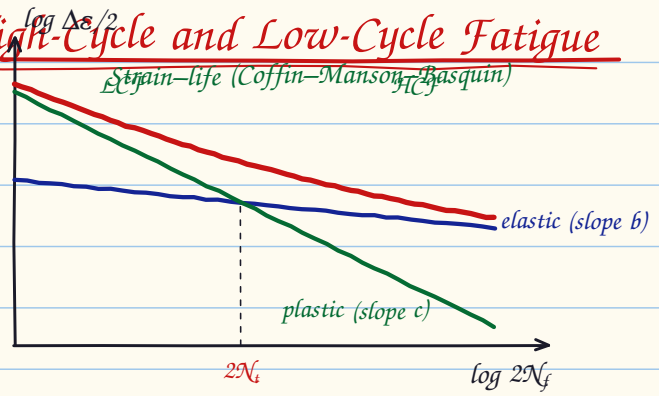
Real components see variable amplitude loads. **Palmgren–Miner linear damage rule** : each cycle at stress level i causes damage $1/N_i$; failure when the sum reaches 1 :

$$D = \Sigma \left(n_i / N_i \right) = 1$$

- n_i = cycles applied at amplitude σ_i, N_i = life at that amplitude from the S–N curve. Remaining life fraction = $1 - D$.
- **Limitations** : ignores load sequence (high-to-low gives $\Sigma < 1$: failure earlier ; low-to-high gives $\Sigma > 1$), neglects interaction effects, notch plasticity, residual stresses, stress below the endurance limit (they do cause damage after cracks start) ; real $\Sigma\ n/N = 0.3–3$. Improved : **Marco–Starkey** ($D = \Sigma\ (n/N)^x$), **Manson's double linear damage rule, Corten–Dolan,** damage curve approaches ; **rainflow counting** (ASTM E1049) to extract cycles from a random history.

Ex 6. A part is run 2×10^4 cycles at σ_1 ($N_1 = 10^5$), 5×10^4 at σ_2 ($N_2 = 5\times10^5$) and 2×10^5 at σ_3 ($N_3 = 10^6$) : $D = 0.2 + 0.1 + 0.2 = 0.5$ ⇒ remaining life 50 % ; if the remainder is at σ_1 : further $n = 0.5 \times 10^5 = 5\times10^4$ cycles.

8. High-Cycle and Low-Cycle Fatigue



$\Delta\varepsilon/2 = (\sigma_f'/E)(2N_f)^b + \varepsilon_f'(2N_f)^c$

$b \approx -0.05$ to -0.12

$c \approx -0.5$ to -0.7

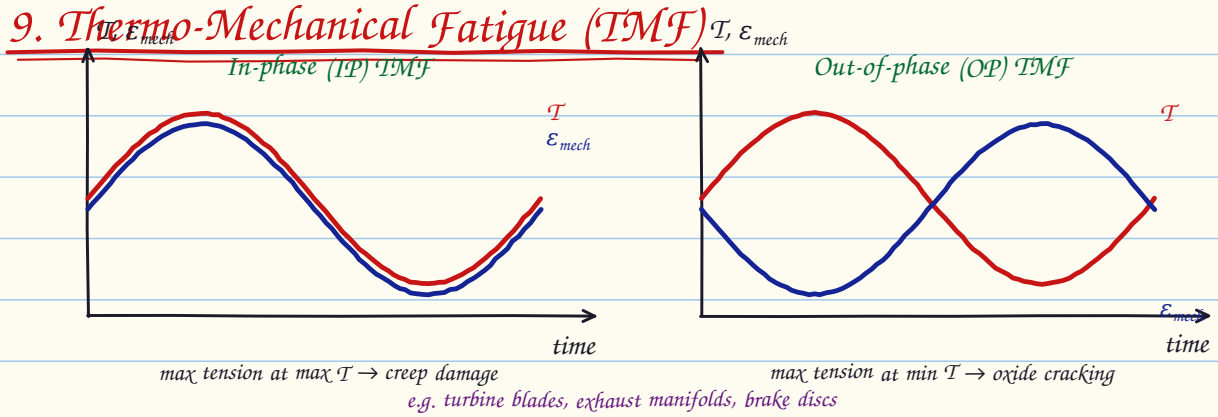
- **HCF** : $N_f > 10^4$–10^5 ; nominal stresses elastic ; stress-controlled – S–N approach ; fatigue life governed by crack initiation ; examples : shafts, springs, bridges, machine parts.
- **LCF** : $N_f < 10^4$–10^5 ; large plastic strain each cycle (local yielding at notches) ; strain-controlled – ε–N approach ; hysteresis loops visible ; crack propagation dominates ; examples : pressure vessel start-up cycles, turbine disc, thermal cycling, earthquake loading.
- **Coffin–Manson** (plastic strain) : $\Delta\varepsilon_p/2 = \varepsilon_f'\ (2N_f)^c$ ($c \approx -0.5$ to -0.7, $\varepsilon_f' \approx$ true fracture ductility). Elastic part (Basquin) : $\Delta\varepsilon_e/2 = (\sigma_f'/E)(2N_f)^b$. **Total strain–life** (Manson–Coffin–Basquin–Morrow) :

$$\Delta\varepsilon/2 = (\sigma_f'/E)(2N_f)^b + \varepsilon_f'\ (2N_f)^c$$

- **Transition life** $2N_t$ where plastic and elastic strain amplitudes are equal ; ductile materials have large $2N_t$ (strain-controlled design), strong brittle materials small $2N_t$ (stress-controlled). **Universal slopes** (Manson) : $\Delta\varepsilon = 3.5\ (S_u/E)\ N^{-0.12} + D^{0.6}\ N^{-0.6}$.

Ex 7. $\varepsilon_f' = 0.5$, $c = -0.6$: a plastic strain amplitude $\Delta\varepsilon_p/2 = 0.005 \Rightarrow (2N)^{-0.6} = 0.01 \Rightarrow 2N = 100^{1/0.6} = 2154 \Rightarrow N = \approx$ 1080 cycles.

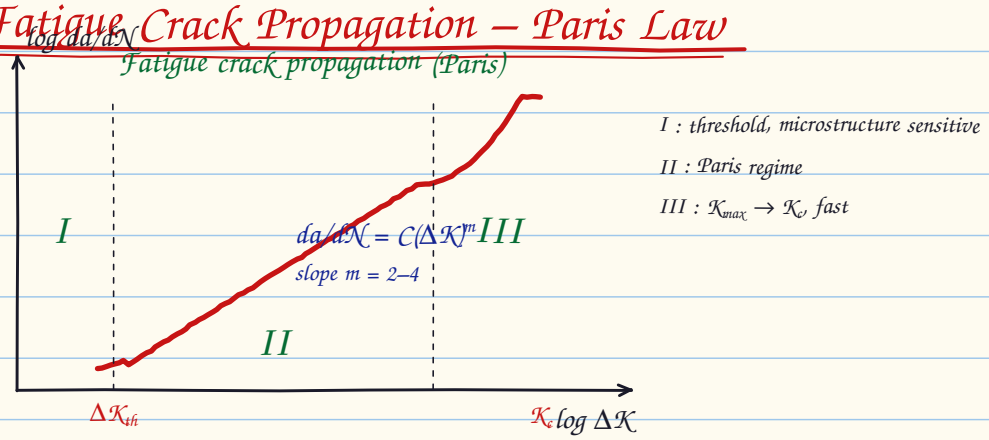
9. Thermo-Mechanical Fatigue (TMF)



TMF : fatigue under simultaneously varying temperature and mechanical strain (not isothermal) – thermal expansion is constrained by the neighbouring material, giving cyclic thermal strain $\varepsilon_{th} = \alpha\,\Delta T$. Total strain $\varepsilon_{mech} = \varepsilon_{total} - \alpha(T - T_0)$.

- Thermal stress for a fully constrained part : $\sigma = E\,\alpha\,\Delta T$ — e.g. steel E = 210 GPa, $\alpha = 12\times10^{-6}/°C$, ΔT = 300 °C ⇒ σ = 756 MPa.
- **Phasing** : in-phase (IP – max tensile strain at max T ⇒ creep damage, ductile) ; out-of-phase (OP – max tension at min T ⇒ oxide-assisted cracking at the surface, more damaging for Ni superalloys, Al). Also diamond cycles, clockwise/counter-clockwise.
- Damage mechanisms : fatigue + creep + oxidation (environmental) interaction ; life models – Neu–Sehitoglu, strain-range partitioning (PP, PC, CP, CC), damage-accumulation, Larson–Miller. Examples : turbine blades/vanes, combustor liners, exhaust manifolds, brake discs, solder joints, nuclear plant piping (thermal shock). Testing : servo-hydraulic machine with induction/resistance heating, extensometer, controlled temperature–strain waveform.
- Reduction : lower thermal gradients, compliant design, materials with low α/E and high conductivity (reduce $E\alpha/k$), thermal barrier coatings.

10. Fatigue Crack Propagation – Paris Law



Fracture mechanics treats the crack growth rate as a function of the stress intensity range $\Delta K = K_{max} - K_{min} = Y \Delta\sigma \sqrt{(\pi a)}$ ($\Delta\sigma$ = tensile range ; compressive part ignored).

- Three regions of the log(da/dN)–log ΔK curve : **I** – below the threshold ΔK_{th} (≈ 2–5 MPa√m for steels ; no growth ; microstructure & R sensitive) ; **II** – linear (Paris) region ; **III** – rapid growth as $K_{max} \to K_c$ (or K_{Ic}) : unstable.

$da/dN = C\,(\Delta K)^m$ (Paris–Erdogan 1963) ; m = 2–4 (steels ≈ 3, Al ≈ 3–4), C from tests (units m/cycle with ΔK in MPa√m)

- **Forman** equation (includes R, region III) : $da/dN = C(\Delta K)^m / [(1 - R)K_c - \Delta K]$; **Walker** : $da/dN = C[\Delta K/(1 - R)^{1-\gamma}]^m$.
- **Fatigue life by integration** from the initial flaw a_i to the critical size a_f :

$N_p = \int_{a_i}^{a_f} da / [\, C\,(Y \Delta\sigma \sqrt{(\pi a)})^m \,] = [\, a_i^{1-m/2} - a_f^{1-m/2} \,] / [\, (m/2 - 1)\, C\, Y^m\, \Delta\sigma^m\, \pi^{m/2} \,]$ ($m \neq 2$, Y const.)

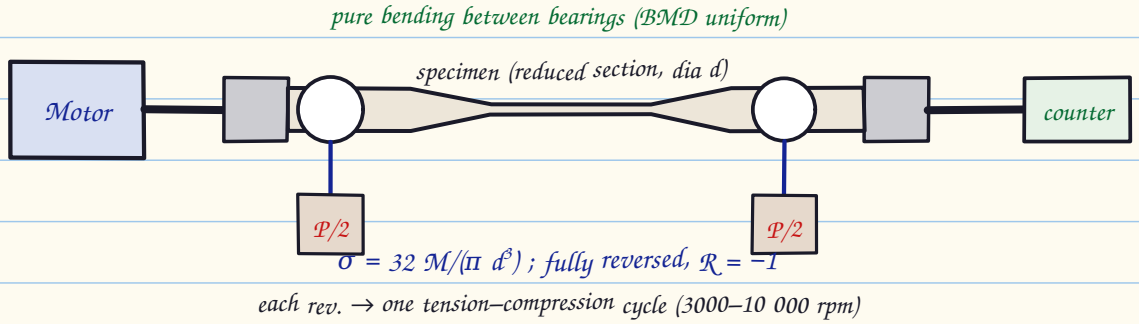
with $a_f = (1/\pi)[K_{Ic}/(Y\,\sigma_{max})]^2$.

Ex 8. $m = 3$, $C = 10^{-11}$ (m/cycle, MPa√m), $Y = 1.12$, $\Delta\sigma = 100$ MPa ($R = 0 \Rightarrow \sigma_{max} = 100$), $K_c = 60$ MPa√m, $a_i = 1$ mm : $a_f = (1/\pi)(60/112)^2 = 91$ mm ; $N_p = [31.6 - 3.31]/[0.5 \times 10^{-11} \times 1.405 \times 10^6 \times 5.568] = 28.3/(3.91\times10^{-5}) = 7.2\times10^5$ cycles. (Most of the life is spent while the crack is small – inspection must detect a_i.)

- Uses : damage-tolerant design (aircraft, rail, pipelines) – set inspection intervals ; retardation by overloads (plastic zone), acceleration by under-loads ; short cracks grow faster than predicted ; effect of mean stress via R. **Increasing fatigue crack growth resistance** : higher K_{Ic}, compressive residual stresses, fine grains (near threshold), avoid corrosive environment.

11. Fatigue Testing Machines

R.R. Moore rotating-beam fatigue machine



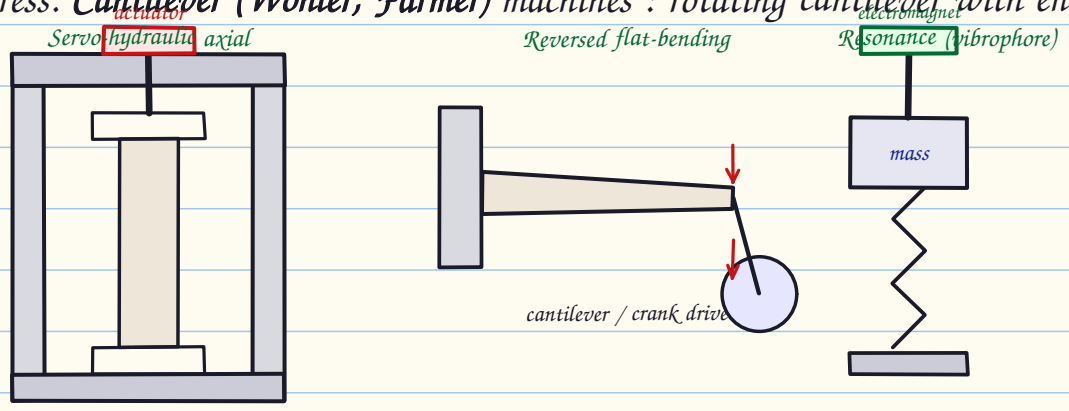
(a) R.R. Moore rotating-beam machine (pure bending)

- A motor-driven, hour-glass specimen is held in two bearings and loaded by weights so that the middle zone sees **pure bending (constant bending moment)** ; as the specimen rotates each fibre experiences tension–compression : fully reversed stress $R = -1$; speed 3000–10 000 rpm ; counter records cycles ; stops automatically at fracture.

$$\sigma = 32\,M / (\pi\, d^3)\ (M = \text{bending moment},\ d = \text{diameter at minimum section})$$

Ex 9. $d = 8$ mm, target $\sigma = 300$ MPa $\Rightarrow M = \sigma\pi d^3/32 = 300\times10^6 \times \pi \times (0.008)^3/32 = 15.1$ N·m (e.g. 151 N at an arm of 0.1 m).

- Cheap, simple, high speed ; limited to reversed bending, small specimens, no mean stress. **Cantilever (Wöhler, Farmer)** machines : rotating cantilever with end load.



(b) Axial (push–pull) servo-hydraulic machine

- Hydraulic actuator + servo-valve + load cell + extensometer under closed-loop digital control (load, strain or displacement waveform – sine, triangle, random, block) ; frequency 0.1–100 Hz ; R any value, mean stress, LCF (strain-controlled),

crack-growth (K-controlled, ASTM E647), TMF with furnace ; Instron, MTS.

(c) Other machines

- **Reversed flat-bending** (sheet – cantilever bent by a crank/eccentric) ; **torsional fatigue** machines ; **resonance (vibrophore) machines** : electro-magnetic or electromechanical excitation at the natural frequency (50–250 Hz) of a spring–mass system – low energy cost, fast, ideal for HCF ; **ultrasonic fatigue** (20 kHz – 10^9–10^{10} cycles in a few days, very high cycle fatigue) ; **rotating-bending with combined axial loads** ; **full-scale component test rigs** (axles, wheels, wings).
- Standards : ASTM E466 (force-controlled axial), E606 (strain-controlled), E647 (crack growth), E739 (statistical analysis), ISO 1099, ISO 12106.
- **Procedure for S–N curve** : 8–12 identical polished specimens (surface finish < 0.2 µm) ; first at $\sigma \approx 0.7\,S_u$, then reduce in steps ; run-out at 10^7 (steel) ; plot log N vs S, draw best fit curve ; for endurance limit use staircase (up-down) method.

12. Additional Solved Problems

P1. A steel shaft sees $\sigma_a = 220$ MPa fully reversed ; $S_u = 700$ MPa, $S_e' = 350$ MPa, $k_a = 0.8$, $k_b = 0.85 \Rightarrow S_e = 350 \times 0.8 \times 0.85 = 238$ MPa $\Rightarrow$ factor of safety $n = 238/220 = 1.08$ – inadequate for design.

P2. Paris-law crack grows from 2 mm to 12 mm at $\Delta\sigma = 80$ MPa, $Y = 1.0$, $m = 2.5$, $C = 5\times10^{-12}$: $a_i^{-0.25} = 4.73$, $a_f^{-0.25} = 3.02 \Rightarrow$ difference 1.71 ; denominator = $0.25 \times 5\times10^{-12} \times 80^{2.5} (= 57\,243) \times \pi^{1.25} (= 4.19) = 3.0\times10^{-7} \Rightarrow N = 1.71/3.0\times10^{-7} = 5.7\times10^6$ cycles.

P3. Goodman : $S_u = 500$, $S_e = 220$ MPa : at $\sigma_m = 150$ MPa, $\sigma_{a,allowed} = 220(1 - 0.3) = 154$ MPa.

P4. $R = 0.1$, $\sigma_{max} = 200$ MPa : $\sigma_{min} = 20 \Rightarrow \sigma_a = 90$ MPa ; $\sigma_m = 110$ MPa ; $A = 0.82$.

Short answers – Quick Revision

- *What is the endurance limit?* Stress amplitude below which a ferrous material has infinite life ($\approx 0.5\,S_u$) ; absent in Al, Cu and corrosive environments.
- *Why does fatigue crack start at the surface?* Maximum stress, free surface slip, stress raisers, environment.
- *What do striations represent?* One per load cycle in Stage II ; spacing ≈ da/dN.

- *Why compressive residual stress is beneficial?* It lowers the effective mean/tensile stress that opens the crack.
- *Difference HCF and LCF?* Elastic stress-controlled > 10^4–10^5 cycles vs plastic strain-controlled < 10^4.

13. Quick Formula Sheet – Unit IV

Item	Formula
Stress cycle	$\sigma_a = (\sigma_{max} - \sigma_{min})/2$; $R = \sigma_{min}/\sigma_{max}$
Basquin	$\sigma_a = \sigma_f' (2N)^b$
Goodman	$\sigma_a/S_e + \sigma_m/S_u = 1$
Marin	$S_e = k_a k_b k_c k_d k_e S_e'$
Notch	$K_f = 1 + q(K_t - 1)$
Miner	$\Sigma\ n_i/N_i = 1$
Strain–life	$\Delta\varepsilon/2 = (\sigma_f'/E)(2N)^b + \varepsilon_f' (2N)^c$
Paris	$da/dN = C(\Delta K)^m$; $\Delta K = Y\Delta\sigma\sqrt{(\pi a)}$
Rotating beam	$\sigma = 32M/\pi d^3$

14. Important Questions

- Define stress cycle terms ; explain S–N curve, endurance limit and fatigue strength. Why do ferrous and non-ferrous metals differ?
- Explain the effect of mean stress (Goodman, Gerber, Soderberg) with a Haigh diagram.
- List and explain the factors affecting fatigue strength ; role of residual stress and surface treatment.
- Describe the stages of fatigue failure and structural changes (PSBs, intrusions, striations, beach marks).
- Explain Miner's rule of cumulative damage with limitations. Differentiate HCF and LCF ; Coffin–Manson relation.
- What is thermo-mechanical fatigue? Explain IP and OP cycles.
- Derive/Explain Paris law and the three regions of crack growth ; compute fatigue life by integration.

- Describe the R.R. Moore rotating-beam machine and axial servo-hydraulic machines.